

Global Change Hydrology (GCH) Summer Training & Practical Program

1–11 SEPTEMBER 2026

Beijing, China

Tsinghua University • Institute of Geographic Sciences and Natural Resources Research, CAS

Program Handbook

1. Welcome and Overview

Welcome to the Global Change Hydrology (GCH) Summer Training & Practical Program, which is sponsored by the Sino-German Center Mobility Program (M-0468): Upscaling processes to improve global water resource models (APPLAUD). The program brings together colleagues from Germany, China and the United Kingdom for technical training, scientific discussion and planning of future collaboration.

Program Theme

Exploring Impacts of Global Change on the Water Systems and Corresponding Mitigation Solutions

Program Focus

Environmental pollutants and forensics; large-scale water quantity and quality modelling; big-data analysis; compound flooding; flood-risk management; and climate adaptation.

Program Objectives

1. Systematically consolidate hydrogeochemical expertise covering water pollutant characteristics, monitoring techniques, and target pollutant analytical methodologies.
2. Equip participants with technical capabilities for large-scale water quantity & quality simulation as well as flood modelling.
3. Develop proficiency in processing large hydrological datasets and applying machine learning techniques to water research.
4. Cultivate scientific thinking and writing and enhance collaborative research capabilities across teams.

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2. Programme

DATE	DAY	MAIN ARRANGEMENT	VENUE
Tue 1 Sep	Arrival & Welcome	Check-in at Wenjin Hotel; rest; welcome ceremony (Kan Lei pickup at airport)	Wenjin Hotel
Wed 2 Sep	Technical Course 1	Properties and partition equilibria of organic compounds in the environment	Tsinghua University Mengminwei South, 821
Thu 3 Sep	Technical Course 2	Chemical analysis of organic compounds & Environmental forensics using microcontaminants	Tsinghua University Mengminwei South, 821
Fri 4 Sep	Research Exchange	Research exchange and laboratory visit (morning) APPLAUD project meeting (afternoon)	Tsinghua University, Mengminwei South, 821
Sat 5 Sep	Weekend	Free time; hosts provide arrangements and assistance	Self-arranged
Sun 6 Sep	Weekend	Free time; hosts provide arrangements and assistance	Self-arranged
Mon 7 Sep	Joint Workshop	Frontiers in water quality research and flood modelling and climate risk research	Tsinghua University Mengminwei South, 818
Tue 8 Sep	Technical Course 3	Large-scale hydrological modelling	IGSNRR-CAS, Room D301
Wed 9 Sep	Technical Course 4	Introduction to large-scale water quality modelling	IGSNRR-CAS, Room D301
Thu 10 Sep	Student Session	Student presentations and faculty feedback; closing ceremony and summary	IGSNRR-CAS, Room D301
Fri 11 Sep	Publication Course 5	Workshop on how to publish scientific papers by Prof. Nigel Wright	IGSNRR-CAS, Room D301
Fri 11 Sep	Departure Day	Hotel check-out and end of programme	Wenjin Hotel

3. Course Schedule — 2/3/8/9/11 September

WED 2 SEP	Properties and Partition Equilibria of Organic Compounds in the Environment Lead(s): Tobias Licha • Venue: Tsinghua University, Mengminwei South Room 821 Core environmental properties, partition behaviour and guided exercises.
09:00–09:05	Host's Opening Remarks (Qihong Tang)
09:05–09:10	Welcome Remarks (Deliang Chen)
09:10–09:15	Welcome Remarks (Martina Flörke)
09:15–12:00	Lecture, concepts and case examples
12:00–13:30	Lunch break
13:30–16:30	Exercises, discussion and practical exchange

THU 3 SEP	Chemical Analysis of Organic Compounds & Environmental Forensics Using Microcontaminants Lead(s): Tobias Licha and Jens Prothmann • Venue: Tsinghua University, Mengminwei South 821 Analytical approaches, pollutant tracing and forensic interpretation.
09:00–12:00	Lecture, concepts and case examples
12:00–13:30	Lunch break
13:30–16:30	Exercises, discussion and practical exchange

TUE 8 SEP	Large-Scale Hydrological Modelling	
	Lead(s): Martina Flörke and Kan Lei • Venue: IGSNRR-CAS, Room D301 Large-scale workflows, data analysis and collaborative exercises.	
TIME	LECTURE / EXERCISE	SPEAKER(S)
09:00–09:10	Survey: Research topics, and experience with global hydrological modelling	Martina Flörke
09:10–09:40	Introduction to the WaterGAP3 Framework	
09:40–10:15	Introduction to the WaterGAP3 Hydrological Model Structure	Martina Flörke, Kan Lei
10:15–10:30	Preparation: WaterGAP3 Installation & Demo Data Download	Kan Lei
10:30–10:45	Coffee Break	
10:45–12:00	Start WaterGAP3 with a Demo Dataset	
12:00–13:30	Lunch Break	
13:30–14:15	Preparation of Topological and Water Body Parameters	Kan Lei
14:15–15:00	Preparation of Physiographical Parameters	
15:00–15:15	Coffee Break	
15:15–16:00	Preprocessing of Meteorological Data and Water Use	
16:00–16:45	Postprocessing: Calibration, Validation, and Visualization	

WED 9 SEP		Introduction to Large-Scale Water Quality Modelling Lead(s): Martina Flörke and Katrin Schweden • Venue: IGSNRR-CAS, Room D301 Model structure, input data, implementation choices and exercises.	
Time	Block	Content (Lecture / Exercise)	Who? / Learning objectives
09:00 – 09:30	Welcome expectations & •	Introductions – name, home university, interest in water quality	• MF, KS • Collection of personal learning goals

			Post-it wall to collect expectations, concerns and questions	and interests
09:30 10:15	–	Lecture 1 – Fundamentals of Water Quality	- Hydrologic cycle & why water quality matters - Key variables (concentration, load, dilution potential) - Typical water-quality parameters (water temperature, bacteria, organics, nutrients)	- MF, KS - Master the basic terminology and recognise the most common water-quality indices
10:15 10:30	–	Coffee break		
10:30 11:15	–	Exercise 1 – Data exploration in R	- Import long-term water temperature time series - Data exploration and quality control (missing values, outliers) - Basic visualisations	- KS, MF - Load raw data, produce simple statistical summaries and plots
11:15 12:00	–	Lecture 2 – Processes in Water Quality	- Transport (advection, dispersion, diffusion) - Reactions (simple introduction) - Sources & sinks (point vs. diffuse, reservoirs, soils, atmosphere)	- MF, KS - Understand the physical and chemical mechanisms that must be represented in a model
12:00 13:30	–	Lunch break		
13:30 14:15	–	Lecture 3 – Model structures & scaling	- WorldQual model, short intro - WorldQual lite model	- MF, KS - Large-scale modelling framework - Given problem, TP loads in a small-/medium-size river basin
14:15 15:00	–	Exercise 2 – Run the WorldQual lite model	Total phosphorus (TP), model input; parameters, method implementation, output	- KS, MF - Carry-out model simulation, baseline 2006-2015 - Model results, visualisation
15:00 15:15	–	Coffee break		
15:15 15:45	–	Lecture 4 – Scenarios	Introduction into scenario assessment	- MF, KS - Familiarize with scenarios
15:45 16:30	–	Exercise 3 – Scenario analysis	Group work; apply WorldQual lite to four plausible future scenarios: (i) climate only, (ii) land use only, (iii) sanitation only, (iv) combined	- KS, MF - Quantify with WorldQual lite future water quality changes (till 2050) - Explore scenarios - Discuss the implications of the changes
16:30 16:55	–	Wrap-up discussion &	- Group feedback: what is clear, what needs more work - Outlook to further modelling tools - Challenges	Consolidate learning, identify remaining gaps
16:55 17:00	–	Farewell		

WED 11 SEP	How to Publish High-Quality Scientific Papers Lead(s): Nigel Wright (Editor-in-Chief, Journal of Flood Risk Management) Venue: IGSNRR-CAS, Room D301 Literature review, discussion, writing, peer-review, revision			
10:00–10:15	Introduction of participants and outline of workshop			
10:15–11:00	Overview of publication process - researcher perspective and editor perspective			
11:00–11:15	Coffee break			
11:15–12:00	What makes a good paper?			
12:00–13:00	Lunch break			
13:00–15:00	1-1 sessions: each student is asked to bring a draft paper in hardcopy to discuss with Prof Wright and get feedback			

4. Research Exchange — 4 September

• Tsinghua University • Mengminwei South Room 821

Scientific presentations and discussion

TIME	TOPIC	PRESENTER
09:00–09:10	Brief introduction	Prof. QiuHong Tang
09:10–09:40	How Vegetation Modulates Local Climate: The Role of Evapotranspiration	Dr. Yitao Li
09:40–10:10	Future flood risk concentration for residents near small rivers across the USA	Dr. Ruijie Jiang
10:10–10:40	Formation mechanisms of persistent extreme precipitation events over the eastern periphery of the Tibetan Plateau: Synoptic conditions and moisture transport.	Dr. Ruiyu Zhao
10:40–11:10	From Canopy to Roots: Representing Vegetation Dynamics for Evapotranspiration Partitioning in the VIC Model	Dr. Dawei Peng
11:10–11:30	Thoughts from Young Professionals: Towards AI Enabled Water Science	Mr. Ying Yan

APPLAUD project meeting

TIME	TOPIC	PRESENTER
14:00–14:10	Brief introduction	Prof. QiuHong Tang
14:10–14:40	Project progress from the Chinese side	Gang Zhao
14:40–15:10	Project progress from the Germany side	Martina Flörke
15:10–16:00	Next step	Martina Flörke and QiuHong Tang

5. Joint Workshop — 7 September

• Tsinghua University • Mengminwei South Room 818

TIME	SESSION / PRESENTATION	SPEAKER(S)	MODERATOR
09:30–09:40	Opening remarks and participant introduction	QiuHong Tang	Xilin Xia
09:40–10:00	Water resources as important factor in the energy transition	Martina Flörke	
10:00–10:20	Global erosion of reservoir flood protection under climate change	Gang Zhao	
10:20–10:40	Influence of high-resolution climate forcing data on global hydrological processes: an analysis using WaterGAP3	Kan Lei	
10:40–11:00	Coffee break (group photo)	--	
11:00–11:40	Investigative monitoring as a component of sustainable groundwater management	Tobias Licha	
11:40–12:00	Large-Scale Water Quality: An Overview of Opportunities and Constraints	Katrin Schweden	
12:00–13:30	Lunch break	All participants	
13:30–13:50	Compound flooding and the role of nature-based solutions in coastal and estuarine systems	Nigel Wright	Gang Zhao
13:50–14:10	Does legacy drainage result in ancient peatland carbon loss through the aquatic pathway?	Paul Gaffney	
14:10–14:30	Global surface water quality gaps for domestic use since 2000	Siao Sun	
14:30–14:50	High-resolution flood modelling and climate risk assessment	Xilin Xia	
14:50–15:10	Coffee break	--	
15:10–15:30	Improving flood monitoring and forecasting via remote sensing for glacier lake outburst floods	QiuHong Tang	
15:30–15:50	Jet stream driven erosivity dipole intensifies asymmetric soil threats over Eurasian black soils	Ximeng Xu	
15:50–16:10	Compound flooding and the role of atmospheric rivers in UK inland catchments	Zifei (Joyce) Jiang	
16:10–16:30	Roundtable: methods, datasets and opportunities for collaboration	All participants	QiuHong Tang

6. Student Session — 10 September

• IGSNRR-CAS • Room D301

TIME	SESSION / PRESENTATION	SPEAKER(S)	MODERATOR
09:30–09:40	Opening remarks and participant introduction	Gang Zhao	Martina Flörke
09:40–10:00	Future climate warming and wetting accelerates Mollisol topsoil depletion by reorganizing freeze-thaw erosion and runoff erosion patterns	Beibei Ding	
10:00–10:20	Water quality constraints across global cities	Yuan Ze	
10:20–10:40	Global irrigation water usage leads to changes in regional available water resources	Rui Jin	
10:40–11:00	Coffee break	—	
11:00–11:20	Global in-situ observation dataset of reservoir water levels and storage volumes: construction, applications, and outlook	Mingyang Zhang	
11:20–11:40	Basic principles of flood response under global change	Yichen Li	
11:40–12:00	Linking surface-water quality to potential disinfection by-product formation	Yixiao Feng	
12:00–14:00	Lunch break	—	
14:00–14:20	Impacts of climate change and reservoir operation on downstream flow regimes in the Omo-Turkana basin, East Africa	Tegegn Takele Mana	Li He
14:20–14:40	Jet stream driven erosivity dipole intensifies asymmetric soil threats over Eurasian black soils	Qianxi Yang	
14:40–15:00	Recent Developments in SiB4 Hydrological Modeling	Wenqian Liang	
15:00–15:10	Coffee break	—	
15:10–15:30	Multi-Source Water-Balance Constraints on Groundwater Exchange in Qinghai Lake	Wenjin Sun	
15:30–15:50	Spatiotemporal Variations and Decoupling Characteristics of Extreme Streamflow and Suspended Sediment Concentration in the Middle Reaches of the Yellow River Basin	Peiwen Shen	Gang Zhao
15:50–16:10	Certificate from instructor(s)	—	
16:10–16:20	Closing remarks (group photo)	Workshop conveners	

7. Participants

Name	Affiliation
Flörke, Martina	Ruhr University Bochum
Lei, Kan (雷坎)	Ruhr University Bochum
Licha, Tobias	Ruhr University Bochum
Prothmann, Jens	Ruhr University Bochum
Schweden, Katrin	Ruhr University Bochum
Jiang, Zifei (Joyce)	University of Birmingham
Wright, Nigel	University of Birmingham
Xia, Xilin	University of Birmingham
Chen, Deliang (陈德亮)	Tsinghua University (清华大学)
Ding, Beibei (丁蓓蓓)	Tsinghua University (清华大学)
Huang, Songjie (黄嵩杰)	Tsinghua University (清华大学)
Jiang, Ruijie (江锐捷)	Tsinghua University (清华大学)
Li, Yitao (李弈韬)	Tsinghua University (清华大学)
Lu, Hui (卢麾)	Tsinghua University (清华大学)
Peng, Dawei	Tsinghua University (清华大学)
Tang, QiuHong (汤秋鸿)	Tsinghua University (清华大学)
Xu, Yan (许琰)	Tsinghua University (清华大学)
Yan, Ying (晏鹰)	Tsinghua University (清华大学)
Zhao, Ruiyu (赵瑞瑜)	Tsinghua University (清华大学)
Paul Gaffney	China University of Geosciences Beijing 中国地质大学(北京)
Li, Yichen (李奕晨)	Zhejiang University (浙江大学)
Feng, Yixiao (冯一效)	中国科学院大学中丹学院
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Liang, Wenqian (梁雯倩)	中国科学院地理科学与资源研究所 (IGSNRR, CAS)
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Shen, Peiwen (沈佩文)	中国科学院地理科学与资源研究所 (IGSNRR, CAS)
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Xu, Ximeng (徐锡蒙)	中国科学院地理科学与资源研究所 (IGSNRR, CAS)
Yang, Qianxi (杨千禧)	中国科学院地理科学与资源研究所 (IGSNRR, CAS)
Yuan, Ze (袁泽)	中国科学院地理科学与资源研究所 (IGSNRR, CAS)
Zhang, Mingyang (张铭洋)	中国科学院地理科学与资源研究所 (IGSNRR, CAS)
Zhao, Gang (赵罡)	中国科学院地理科学与资源研究所 (IGSNRR, CAS)

8. Practical Information

Flight

- September 1st 11:15AM arrived at the Beijing Capitol Airport (4 person, Kan arrived on August 29th).
- September 11th 11:00AM departure from Beijing Capitol Airport (4 person, Kan will stay a bit longer in China)

Dinning

DATE	LUNCH	DINNER
Sep 2 Wed Sep 3 Thu Sep 4 Fri	Meal voucher Tsinghua University 芝兰园食堂 (Zhilan Garden Canteen)	Meal voucher Tsinghua University 芝兰园食堂 (Zhilan Garden Canteen)
Sep 5 Sat Sep 6 Sun	Flexible (with receipt)	Flexible (with receipt)
Sep 7 Mon	Meal voucher Tsinghua University 芝兰园食堂 (Zhilan Garden Canteen)	Group dinner
Sep 8 Tue Sep 9 Wed	Meal voucher buffet 北餐厅地下一层 (North Canteen, IGSNRR)	Meal voucher buffet 北餐厅地下一层 (North Canteen, IGSNRR)
Sep 10 Thu	Meal voucher buffet 北餐厅地下一层 (North Canteen, IGSNRR)	Group dinner
Sep 11 Fri	Meal voucher buffet 北餐厅地下一层 (North Canteen, IGSNRR)	Meal voucher buffet 北餐厅地下一层 (North Canteen, IGSNRR)

Accommodation

ITEM	DETAIL (Germany)	DETAIL (UK)
Hotel	Wenjin Hotel, Beijing (文津国际酒店)	Dequan Hotel, Beijing (德全酒店)
Check-in	Tuesday 1 September 2026	Monday 7 September 2026 (booked from 6th)
Check-out	Friday 11 September 2026	Sunday 13 September 2026
Booking note	Five rooms for the German delegation	One room for UK participant

Host Venues

VENUE	ADDRESS
Tsinghua University	30 Shuangqing Road, Haidian District, Beijing • Courses on Sep 2–4 and workshop on Sep 7
IGSNRR-CAS	11A Datun Road, Chaoyang District, Beijing • Courses on Sep 8–11

Contact

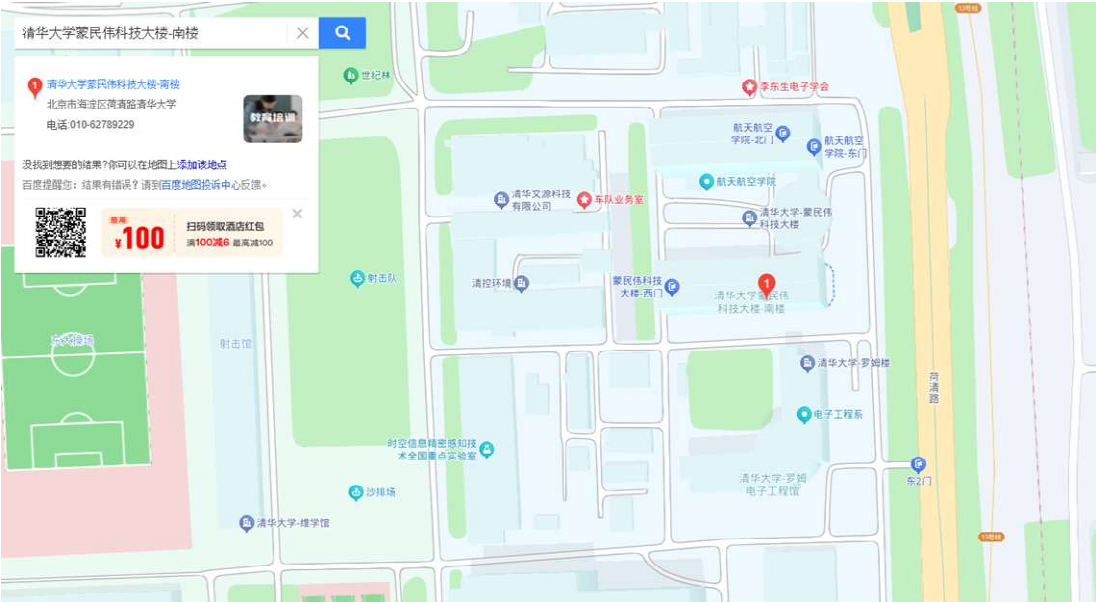
- Gang Zhao, IGSNRR-CAS • gzhao@igsnrr.ac.cn, 18836115276
- Yan Xu, THU • xuyan@tsinghua.edu.cn, 13011815067

9. Location and Map

Wenjin Hotel （文津国际酒店）



Mengminwei Buiding, Tsinghua University (蒙民伟科技大楼南楼 821/818)



IGSNRR, CAS Room D301 （中国科学院地理科学与资源研究所 D 座 301）

